Thevathayarajh Thayananthan

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RESEARCH FOCUS

Research focuses on AI-enabled autonomous systems for agriculture, integrating computer vision, multimodal sensing, 3D localization, autonomous field navigation, and robotic manipulation for selective harvesting and precision crop operations. Develops complete perception-to-action robotic systems, from ROS and Gazebo simulation to integrated hardware and field deployment, to advance reliable, practical, and affordable agricultural automation.

EDUCATION

Ph.D., Agricultural Engineering — Aug 2025 – Present
University of Georgia, Athens, GA, USA
Advanced to candidacy August 2026; **Expected graduation May 2027**
Dissertation Proposal: *Design, Development, Integration and Field Evaluation of a Robotic Cotton Picker*

Doctoral coursework and research, Biosystems Engineering — Jan 2023 – Jul 2025
Mississippi State University, Mississippi State, MS, USA
Program transferred to the University of Georgia with advisor

M.Sc., Electrical and Electronics Engineering — Jan 2019 – Jul 2022
University of Peradeniya, Sri Lanka
Project: IoT-based medical instrumentation for remote patient monitoring

B.Tech. (Hons.), Science and Technology — Mechatronics — Apr 2014 – Jan 2018
Uva Wellassa University of Sri Lanka, Badulla, Sri Lanka
Dissertation: *Development of a monitoring device for the fermentation stage of black tea manufacturing*

RESEARCH APPOINTMENTS

Graduate Research Assistant — Aug 2025 – Present
Sensing and Automation in Agri-Systems (SAAS) Lab, School of Environmental, Civil, Agricultural & Mechanical Engineering, University of Georgia

- Develop perception-driven manipulator control in ROS and Python for a dual-side robotic cotton picker. Completed field testing with one ZED 2i stereo camera, a UR5e manipulator, and a custom end-effector, and developed a two-camera control pipeline for dual-side harvesting.
- Benchmark 33 YOLO variants (v8–v13) and several segmentation models on a 1,008-image, four-class annotated field dataset, comparing mAP and inference speed on matched test splits.
- Developed **CottonID**, a full-stack web application using Next.js, FastAPI, and SQLite to generate boll counts, yield estimates, and defoliation-timing recommendations. Currently improving the underlying perception models; deployed through Cloudflare Tunnel for demonstrations.
- Manage equipment, operations, data infrastructure, and safety within the Sensing & Automation in Agri-Systems (SAAS) Lab.

Graduate Research Assistant — Jan 2023 – Jul 2025
Department of Agricultural and Biological Engineering, Mississippi State University

- Designed and open-sourced **CottonSim**, a ROS/Gazebo simulation environment integrating a virtual cotton farm, boll perception, and vision-guided navigation. The Cotton-Eye module (YOLOv8n-seg) reached 85.2% mAP, 88.9% recall, and

93.0% precision; navigation achieved 100% GPS-guided and 96.7% map-based traversal success within 0.25 m [J1].

- Developed an in-field multi-ripeness blackberry-detection pipeline for soft robotic harvesting from 1,086 images; benchmarking nine YOLO architectures, where YOLOv7-base produced 91.1% mAP overall, 92.4% for ripe berries, and 12.6 ms inference per 1024×1024 image [J2].
- Developed a semantic-segmentation pipeline for automating bladeless catfish cutting, benchmarking five architectures across five classes on 396 images; SegFormer-B5 reached 89.2% mIoU and 94.6% mean pixel accuracy [C3].
- Named as an inventor on a university technology disclosure for an autonomous vision-guided cotton-harvesting system.

Research and Development Intern Sep 2017 – Feb 2018
Research & Innovation Department, MAS Intimates (Unichela Pvt. Ltd., Vidiyal Division), Sri Lanka

- Contributed to seven industrial-automation projects in garment manufacturing, including PLC-based finished-goods scanning and garment folding, an Arduino-based chiller-shutdown SMS alert system, pedal-free and automated sewing, and waistband cutting with mark elimination.

ACADEMIC APPOINTMENTS

Lecturer (Full-time) Mar 2019 – Dec 2022
Department of Science and Technology, Faculty of Applied Sciences, Uva Wellassa University of Sri Lanka

- Instructor of record for eight distinct undergraduate courses totaling 18 credit hours across four academic years (see Teaching)
- Designed curricula for seven new undergraduate courses for the B.Sc. in Mechatronics curriculum revision, including Computer Vision and IoT Development.
- Supervised 15 undergraduate final-year research projects, twelve of which produced peer-reviewed conference abstracts/papers [U1]–[U12].

Demonstrator (Temporary) Mar 2018 – Mar 2019
Department of Science and Technology, Faculty of Applied Sciences, Uva Wellassa University of Sri Lanka

- Led undergraduate laboratory and practical sessions in electronics, instrumentation, and workshop practice, supervising groups of students in the lab class and grading laboratory reports.
- Maintained teaching laboratory equipment and assisted faculty with demonstrations, practical assessments, and student project supervision.

PUBLICATIONS

Name in **bold**; *corresponding author. Earlier work appears under the name **Thevathayarajh, T.** Citations: 41 | h-index: 4 (Google Scholar).

Peer-Reviewed Journal Articles

[J1] **Thayananthan, T.**, Zhang, X.*, Huang, Y., Chen, J., Wijewardane, N.K., Martins, V.S., Chesser, G.D., and Goodin, C.T. (2025). CottonSim: A vision-guided autonomous robotic system for cotton harvesting in Gazebo simulation. *Computers and Electronics in Agriculture*, 239, 110963. `https://doi.org/10.1016/j.compag.2025.110963`

[J2] **Thayananthan, T.**, Zhang, X.*, Harjono, J., Huang, Y., Liu, W., McWhirt, A.L., Threlfall, R.T., and Chen, Y. (2025). In-field multi-ripeness blackberry detection for soft robotic harvesting. *Journal of the ASABE*, 68(6), 1073–1089. `https://doi.org/10.13031/ja.16300`

Book Chapters

[B1] Gunaratnam, A., **Thayananthan, T.**, Thangathurai, K., and Abhiram, B. (2024). Computer vision in livestock management and production. In *Engineering Applications in Livestock Production*, pp. 93–128. Academic Press. https://doi.org/10.1016/B978-0-323-98385-3.00002-5

Peer-Reviewed Conference Proceedings

[C1] **Thayananthan, T.** and Zhang, X.* (2025). Perception-enabled manipulator control for a robotic cotton picker with dual-side harvesting capability. *IFAC-PapersOnLine*, 59(23), 332–337. (8th IFAC Conference on Sensing, Control and Automation Technologies for Agriculture, AGRICONTROL 2025, UC Davis, CA, USA.) https://doi.org/10.1016/j.ifacol.2025.11.809

[C2] Vimalambikaipakan, G., Amarasinghe, C., Rajapaksha, T., **Thayananthan, T.**, and Mariyathas, J. (2024). Traffic sign recognition and auditory alert system for Sri Lankan drivers using deep learning. In *Proc. International Research Conference on Smart Computing and Systems Engineering (SCSE)*, 7, 1–5. IEEE. https://doi.org/10.1109/SCSE61872.2024.10550615

[C3] **Thayananthan, T.**, Zhang, X.*, Liu, W., Yao, T., Huang, Y., Wijewardane, N.K., and Lu, Y. (2023). Automating catfish cutting process using deep learning-based semantic segmentation. In *Sensing for Agriculture and Food Quality and Safety XV*, 12545, 103–116. SPIE. https://doi.org/10.1117/12.2663370

[C4] Zhang, X.*, **Thayananthan, T.**, Usman, M., Liu, W., and Chen, Y. (2023). Multi-ripeness level blackberry detection using YOLOv7 for soft robotic harvesting. In *Autonomous Air and Ground Sensing Systems for Agricultural Optimization and Phenotyping VIII*, 12539, 85–96. SPIE. **Best Paper Award.** https://doi.org/10.1117/12.2663367

[C5] Yapa, A., Ratnayake, K., Premarathna, C.P., and **Thayananthan, T.** (2023). Leveraging virtual reality for robot manipulator education. In *Proc. Moratuwa Engineering Research Conference (MERCon)*, 544–549. IEEE. https://doi.org/10.1109/MERCon60487.2023.10355455

[C6] **Thayananthan, T.**, Wanniarachchi, D.D.C., and Kumari, K.W.S.N. (2019). A digital device to monitor the colour changes during the fermentation stage of black tea processing. In *Proc. International Conference on Recent Advances in Computer Science and Engineering (IC-RACE)*, 75–79.

Preprints and Manuscripts in Preparation or Review

[S1] **Thayananthan, T.**, Zhang, X.*, Laddusinghe Badu, I., Harjono, J., Rains, G.C., Li, B., Bastos, L.M., Wijewardane, N.K., and Martins, V.S. (2026). Selective cotton boll localization for robotic harvesting: evaluation of deep learning vision models under field conditions. *arXiv preprint* arXiv:2609.19592. Under preparation for journal submission. https://arxiv.org/abs/2609.19592

[S2] **Thayananthan, T.** and Zhang, X. Toward fully autonomous cotton harvesting robots: a systematic review of perception, navigation, path planning, and harvesting mechanisms. *Under internal review* (2026).

[S3] **Thayananthan, T.** and Zhang, X. Field integration and evaluation of an autonomous robotic cotton picker. *In preparation* (2026).

Supervised Undergraduate Research — Conference Abstracts/Papers

[U1] Ahamed, M.M.R., Amarasinghe, C., **Thayananthan, T.**, De Silva, D.S.P.P.G., De Zoysa, K., and De Silva, D.P.N. (2023). Development of automated image capturing for zebrafish embryo toxicity model. *Proc. International Research Conference, Uva Wellassa University.*

[U2] Geerthana, V., **Thayananthan, T.**, Amarasinghe, A.R.P.C.C.J., Janotheepan, M., and Lakmali, R.M.T. (2023). Deep learning-based traffic sign recognition and auditory alert system for Sri Lankan drivers. *Proc. International Research Conference, Uva Wellassa University.*

[U3] Liyadipita, L.A.M.G.S.K., Amarasingha, A.R.P.C.C.J., **Thayananthan, T.**, and Bandara, T.U.K.S. (2023). IoT-based patient monitoring system for Sri Lanka. *Proc. International Research Conference, Uva Wellassa University.*

[U4] Jayawardhana, D.G.H., **Thayananthan, T.**, Ekanayake, R.M.T.C.B., and Senanayake, D.D.B. (2022). Automatic billing system for a Sri Lankan pastry shop. *Proc. International Research Conference, Uva Wellassa University.*

[U5] Alwis, D.M.G.P., **Thayananthan, T.**, Ekanayake, R.M.T.C.B., and Senanayake, D.D.B. (2022). Motorbike assistance tool using image processing techniques. *Proc. International Research Conference, Uva Wellassa University.*

[U6] Sampath, W.P.A.C., **Thayananthan, T.**, Prematilake, M., and Ekanayake, R.M.T.C.B. (2022). Develop-

ment of an IoT-based automated colony counter. *Proc. International Research Conference, Uva Wellassa University.*

[U7] Arawa, A.M.P.R.B., **Thayananthan, T.**, Mehendran, Y., and Senanayake, D.B.B. (2022). Recognition of Sinhala machine-printed text for postal address interpretation and postal automation. *Proc. International Research Conference, Uva Wellassa University.*

[U8] Dissanayaka, D.M.K.C., Vidanapathirana, A.C., and **Thayananthan, T.** (2022). Remote monitoring and controlling of an automobile system using NodeMCU. *Proc. International Research Conference, Uva Wellassa University.*

[U9] Gayashan, S.A.V., Vidanapathirana, A.C., Ekanayake, R.M.T.C.B., and **Thayananthan, T.** (2022). Development of a wind turbine system for electric and hybrid carts. *Proc. International Research Conference, Uva Wellassa University.*

[U10] Srikaran, K. and **Thayananthan, T.** (2021). Development of an IoT-based tree-cutting warning system in forests. *Proc. 3rd South Asia Conference on Interdisciplinary Research*, 14.

[U11] Danijal, T.J. and **Thayananthan, T.** (2021). Development of a tracking over-speed system using IoT technology for vehicles. *Proc. International Research Conference, Uva Wellassa University*, 139.

[U12] Senanayake, D.D.B., Kumari, K.W.S.N., and **Thayananthan, T.** (2020). Active and passive safety system for differently abled people and adults. *Proc. International Research Conference, Uva Wellassa University*, 195.

[U13] **Thayananthan, T.**, Amarasinghe, A.R.P.C.C.J., and Ekanayake, R.M.T.C.B. (2019). A digital device to measure distance on multiple surfaces. *Proc. International Research Conference, Uva Wellassa University*, 504.

CONFERENCE PRESENTATIONS

[P1] **Thayananthan, T.** (2026, July). Develop and evaluate a simplified end-effector for robotic selective cotton harvesting. ASABE Annual International Meeting, Indianapolis, IN, USA.

[P2] **Thayananthan, T.** (2026, March). Harvesting cotton before it's too late: a smarter robotic approach. UGA Institute for Integrative Precision Agriculture Spring Retreat, Griffin, GA, USA. **Third Place, Three Minute Thesis oral presentation.**

[P3] **Thayananthan, T.** (2025, August). Perception-enabled manipulator control for a robotic cotton picker with dual-side harvesting capability. 8th IFAC Conference on Sensing, Control and Automation Technologies for Agriculture (AGRICONTROL 2025), Davis, CA, USA.

[P4] **Thayananthan, T.** (2025, April). Cotton-SAM: AI-driven cotton boll localization for autonomous picking using YOLO and SAM integration. AI in Agriculture and Natural Resources Conference, Starkville, MS, USA.

[P5] **Thayananthan, T.** (2024, October). Cotton boll localization system to enable autonomous cotton picking using YOLO and SAM. Graduate Student Research Symposium, Mississippi State University, Mississippi State, MS, USA.

[P6] **Thayananthan, T.** (2024, July). Development of a cotton boll detection system to enhance autonomous picking using YOLOv8 and SAM. ASABE Annual International Meeting, Anaheim, CA, USA.

[P7] **Thayananthan, T.** (2024, July). Computer vision-enabled autonomous robotic navigation system for cotton farms in a Gazebo simulation environment. ASABE Annual International Meeting, Anaheim, CA, USA.

[P8] **Thayananthan, T.** (2024, March). Which is better cotton picker? Three Minute Thesis presentation, Mississippi Agriculture Consortium, Mississippi State University, Starkville, MS, USA.

[P9] **Thayananthan, T.** (2023, July). Catfish cutting automation through computer vision and deep learning. Mississippi Academy of Sciences 5th Summer Science & Engineering Symposium, Mississippi State, MS, USA.

[P10] **Thayananthan, T.** (2023, May). Automating catfish cutting process using deep learning-based semantic segmentation. *Sensing for Agriculture and Food Quality and Safety XV* session, SPIE Defense + Commercial Sensing, Orlando, FL, USA.

[P11] **Thayananthan, T.** (2023, May). Multi-ripeness level blackberry detection using YOLOv7 for soft robotic harvesting. *Autonomous Air and Ground Sensing Systems for Agricultural Optimization and Phenotyping VIII* session, SPIE Defense + Commercial Sensing, Orlando, FL, USA. **Best Paper Award.**

[P12] **Thayananthan, T.** (2018, February). Development of a monitoring device for the fermentation stage of

black tea manufacturing. *Mechanical Engineering and Mechatronics* session, 2nd International Research Symposium, Uva Wellassa University, Badulla, Sri Lanka.

HONORS AND AWARDS

- **Third Place, Three Minute Thesis (3MT) Competition** — Institute for Integrative Precision Agriculture Retreat, University of Georgia (2026). Includes a $500 conference travel award.
- **Hall of Fame Awardee**, College of Engineering — The Graduate School, Mississippi State University (2025)
- **SEC Emerging Scholar** — Selected to represent Mississippi State University in the Southeastern Conference Emerging Scholars Program (2023–2024)
- **Best Paper Award** — Autonomous Air and Ground Sensing Systems for Agricultural Optimization and Phenotyping VIII, SPIE (2023)

FUNDING AND INTELLECTUAL PROPERTY

Intellectual Property

- **Technology Disclosure** — "Design and development of an autonomous vision-guided cotton picker for selective and targeted robotic picking." Mississippi State University Office of Technology Management (2025). Co-inventor with Dr. Xin Zhang.
  `msstate-innovations.technologypublisher.com/technology/58899`

Competitive Travel Awards

- CENGR Graduate Student Travel Grant, College of Engineering, University of Georgia (2026) — $735
- IIPA 3MT Conference Travel Award, University of Georgia (2026) — $500
- TAGGS Travel Scholarship, The Graduate School, Mississippi State University (2024) — $700
- AG2PI Travel Award, Agricultural Genome to Phenome Initiative (2024) — $1,500

Proposal Experience

- Southern SARE Graduate Student Grant (2024) — $22,000; submitted, not funded.

TEACHING EXPERIENCE

Instructor of Record — Uva Wellassa University of Sri Lanka (Mar 2019 – Dec 2022)

Sole instructor for eight undergraduate courses (18 credit hours total): lectures, laboratory design, assessment, and grading.

- **SCT 364-2 Embedded Systems** (2 credits) — 2019, 2020, 2021, 2022
- **SCT 462-2 Intelligent Systems** (2 credits) — 2019, 2020, 2021, 2022
- **SCT 319-2 Maintenance, Production, and Project Management** (2 credits) — 2021, 2022
- **SCT 363-2 Electrical Systems** (2 credits) — 2020, 2021
- **SCT 408-2 Industrial Electronics** (2 credits) — 2020, 2021
- **SCT 362-3 Digital and Analog Electronics** (3 credits) — 2020
- **SCT 373-2 Electric Power and Machines** (2 credits) — 2019
- **SCT 142-3 Engineering Workshop** (3 credits) — 2019

Course Development

BSc Mechatronics curriculum revision, Department of Science and Technology, Uva Wellassa University (2019). Invited by the department to design seven new course modules and their lesson plans:

- MEC 361-2 Computer Vision
- MEC 331-2 IoT Development
- MEC 232-2 Embedded Systems
- MEC 233-2 Embedded Systems Design
- MEC 362-2 Transducers and Instrumentation
- MEC 111-2 Computer Programming I
- MEC 212-2 Computer Programming II

Course Coordination and Laboratory Instruction

- **Graduate Teaching Assistant**, ABE 4423/6423 Bioinstrumentation II, Mississippi State University (Spring 2024). Laboratory instructor with Dr. Xin Zhang. Certified Teaching Assistant, Level 2/3.
- **Subject Coordinator**, BGE 121-2 Ethics and Law Basics, Uva Wellassa University (2022). Cross-faculty course delivered to three faculties; coordinated visiting resource persons, assessment, and results submission.
- **Coordinator**, Short Course in Robotics and Arduino Programming, Centre for Open and Distance Learning, Uva Wellassa University (2019–2022). Appointed by the Vice Chancellor; responsible for curriculum, timetabling, certification, and course budget.

Courses Prepared to Teach

Computer Vision for Agricultural and Biological Systems; Artificial Intelligence for Agricultural and Biological Engineering; Machine Learning Applications in Agriculture; Agricultural Automation and Precision Agriculture.

MENTORING AND ADVISING

Undergraduate and Visiting Researchers — six students mentored, Sensing and Automation in Agri-Systems (SAAS) Lab (2024 – Present)

- **Isadora Goncalves** (Aug 2026 – present) — School of Electrical & Computer Engineering, University of Georgia. End-effector task design for a robotics competition platform.
- **Jap Ja Mun Pan** (May – Jul 2026) — Electrical & Computer Engineering, Trevecca Nazarene University; visiting researcher through the Summer Undergraduate Research Opportunities in Engineering (SUROE) program, University of Georgia. Actuation design for a plant surgical robot.
- **Madison Mateyko** (Nov – Dec 2025) — School of Environmental, Civil, Agricultural & Mechanical Engineering, University of Georgia. Computer vision for the robotic cotton picker.
- **Riley Gebke** (Mar – May 2025) — Department of Agricultural & Biological Engineering, Mississippi State University. Computer vision for the robotic cotton picker.
- **Mary J. Echelberry** (Oct – Dec 2024) — Department of Agricultural & Biological Engineering, Mississippi State University. Computer vision for the robotic cotton picker.
- **Kelly Bilington** (Apr 2024) — Harper Adams University, United Kingdom; graduate student worker through the Mississippi State University–Harper Adams University Student Exchange Program, hosted by the Agricultural Autonomy Institute. Development of a UR5e manipulator base platform for integration with a Husky A200 mobile robot.

Undergraduate Final-Year Research Projects — Uva Wellassa University (Mar 2019 – Dec 2022). Supervised 15 projects; twelve produced peer-reviewed conference abstracts [U1]–[U12]. Topics spanned computer vision, deep learning, IoT systems, and mechatronic design, including traffic sign recognition, zebrafish embryo toxicity imaging, automated microbial colony counting, Sinhala postal-address OCR, and vehicle safety systems.

Student Support Roles — Uva Wellassa University

- **Student Counselor**, Faculty of Applied Sciences (2022–2023). Completed a 40-hour Psychological Counselling Program.
- **Academic Sub-Warden**, University residence halls (2019–2023). Residential academic support and student welfare.
- **Student Mentor**, Faculty of Applied Sciences (2019–2023). Academic guidance for undergraduate students.

PROFESSIONAL SERVICE

Editorial Board Service

- **Associate Editor, Computer Science and Engineering** — *Endeavors: Mississippi State Undergraduate Research Journal* (ISSN 3071-012X), Mississippi State University (2024–2025). Inaugural editorial board of MSU's first peer-reviewed undergraduate research journal; conducted double-blind peer review of engineering and computer science submissions for Volume 1.
  `scholarsjunction.msstate.edu/endeavors`

Journal Peer Review — 17 completed reviews across nine journals

- *Computers and Electronics in Agriculture*, Elsevier (6 reviews; 2024–present)
- *The International Journal of Robotics Research*, SAGE (1 review; 2026)

- *Scientific Reports*, Nature Portfolio (1 review; 2026)
- *Journal of Food Composition and Analysis*, Elsevier (2 reviews; 2026)
- *Frontiers in Plant Science*, Frontiers (3 reviews; 2026)
- *Food and Energy Security*, Wiley (1 review; 2026)
- *Food and Humanity*, Elsevier (1 review; 2026)
- *Journal of Atmospheric and Solar-Terrestrial Physics*, Elsevier (1 review; 2026)
- *Endeavors: Mississippi State Undergraduate Research Journal*, Mississippi State University (1 review; 2025)

Conference Service

- Track Coordinator, Engineering and Technology track, International Research Conference, Uva Wellassa University (2022)
- Reviewer, International Research Conference, Uva Wellassa University (2021–2022)
- Reviewer, 2nd International Conference on Advancements in Computing, Sri Lanka Institute of Information Technology (2021)
- Session Panel Member, International Research Conference, Uva Wellassa University (2020)
- Organizing Committee Member, International Research Conference, Uva Wellassa University (2019, 2020)

INSTITUTIONAL SERVICE AND GOVERNANCE

- **Lab Manager**, SAAS Lab — Mississippi State University (2024–2025); University of Georgia (2025–present)
- **Member, Faculty Board**, Faculty of Applied Sciences, Uva Wellassa University (2022–2023). Appointed by the Dean under S.48(1A)(c) of the Universities Act No. 16 of 1978. The Faculty Board is the governing academic body for curricula, examinations, and policy. Three-year term, curtailed on departure for doctoral study.
- **Member**, Program Review and Self-Evaluation Report (SER) groups 3, 4, and 8, Uva Wellassa University (2022–2023)
- **Member**, Graduate Employability Survey team, AHEAD ELTA-ELSE Faculty Development Plan, Uva Wellassa University (2022–2023). Appointed by the Dean; data informed the final AHEAD project report.
- **Faculty Advisor**, IEEE Robotics and Automation Society Student Chapter, Uva Wellassa University (2021–2023)
- **Panel Member**, Stakeholder Meeting for the BSc Mechatronics Curriculum Revision, Uva Wellassa University (2020)
- **Co-author**, Survey of Postgraduate Admission Pathways followed by UWU graduates (2020)
- **Member (Activity 02)**, AHEAD Project — Revising the Curricula of Existing Degree Programmes to Meet Local and International Demands, Uva Wellassa University (2019–2020)
- **Organizing Committee Member**, Convocation, Uva Wellassa University (2019)

OUTREACH AND COMMUNITY ENGAGEMENT

- **Secretary**, Softball Cricket Club, Mississippi State University (2024–2025)
- **Social Secretary**, Softball Cricket Club, Mississippi State University (2023–2024)
- **Webmaster**, Sri Lankan Association, Mississippi State University (2023–2024)
- **Workshop Instructor**, Engineering Machinery Workshop for School Students, Uva Wellassa University (2018)
- **Organizing Committee**, UWU Robot Battles, Department of Science & Technology, Uva Wellassa University (2016)
- **Volunteer Mathematics Instructor**, G.C.E. Ordinary Level (O/L) Seminar Series, Uva Wellassa University, Sri Lanka (2014–2017). Conducted mathematics seminars for students at resource-constrained schools with limited access to supplemental educational support, with the goal of improving examination preparedness, pass rates, and student interest in mathematics.

PROFESSIONAL DEVELOPMENT

- **Peer Review Certification** — Elsevier Researcher Academy (2026): Fundamentals of Peer Review; Certified Peer Reviewer Course; Becoming a Peer Reviewer (14 hours total)
- **Preparing Future Faculty Program**, Mississippi State University (2024–2025)
- **Certified Teaching Assistant, Level 2/3**, Mississippi State University (2024)
- **Fundamentals of Deep Learning**, Certificate of Competency, NVIDIA Deep Learning Institute (2023)
- **Psychological Counselling Program** (40 hours), Staff Development Centre, Uva Wellassa University / World Bank AHEAD Project (2021–2022)
- **Workshop: Writing Manuscripts and Proposals for Research Grants**, AHEAD ELTA-ELSE Faculty Development Plan, Uva Wellassa University (2020)
- **Workshop: Online Teaching Tools and Methods for Effective Teaching and Assessment**, Uva Wellassa University (2020)
- **Certificate Course in Teaching in Higher Education**, Staff Development Centre, Uva Wellassa University (2019)

TECHNICAL SKILLS

Programming: Python, C++, MATLAB
Deep learning: PyTorch, TensorFlow/Keras, YOLO (v5–v26), SegFormer, instance and semantic segmentation, dataset design and annotation
Robotics and perception: ROS, Gazebo, OpenCV, RGB-D and stereo vision (ZED 2i, RealSense), UR5e manipulator control, Clearpath Husky, GPS and map-based autonomous navigation, SLAM
Hardware and embedded: Arduino, PLC, microcontroller systems, IoT sensor networks, instrumentation design

PROFESSIONAL MEMBERSHIPS

- Integrative Precision Agriculture Student Club (IPAC), University of Georgia, since 2025
- American Society of Agricultural and Biological Engineers (ASABE), Graduate Student Member, since 2024
- SPIE — the International Society for Optics and Photonics, Student Member, since 2023
- Institute of Electrical and Electronics Engineers (IEEE), Graduate Student Member, since 2020

RESEARCH SOFTWARE AND DATA

- **CottonID** (2026) — web application for cotton boll detection, yield estimation, and defoliation-timing decision support (Next.js, FastAPI, RF-DETR/YOLO).
- **CottonBoll Harvest** (2026) — supplementary code, trained-model resources, and supporting materials for the manuscript *Selective Cotton Boll Localization for Robotic Harvesting: Evaluation of Deep Learning Vision Models under Field Conditions*. Repository private during manuscript preparation. Public release planned upon publication.
  `github.com/imtheva/CottonBoll_Harvest`
- **CottonSim** (2025) — open-source ROS/Gazebo simulator, virtual cotton farm, and vision-guided navigation stack for autonomous cotton picking.
  `github.com/imtheva/CottonSim`
- **Multi-Ripeness Blackberry** (2025) — detection models and annotated field dataset (1,086 images, three ripeness classes, two seasons).
  `github.com/Zhanglab-abe/Multi-Ripeness_Blackberry`
- **Catfish-Segment** (2023) — semantic segmentation models for automated catfish processing.
  `github.com/Zhanglab-abe/Catfish-Segment`

MEDIA COVERAGE

- MSU Innovations, "Design and development of an autonomous vision-guided cotton picker for selective and targeted robotic picking" (October 2025)
- Farm Flavor, "Agriculture technology advancements revolutionize Mississippi ag industry" (November 2024)
- WCBI News, "Mississippi State researchers develop prototype to advance cotton industry" (November 2023)
- MSU Graduate School Newsletter, award announcement (October 2023)